



Chapter 3 Fundamentals of Fluid Motion

Section 1 How to describe fluid motion

Section 2 Basic concepts of fluid flow

Section 3 Analysis of fluid particle motion

Section 4 Basic equations of fluid flow

Section 5 Integration of Euler equations of motion

Section 6 Steady, 2D, potential, incompressible flow



Chapter 3 Fundamentals of Fluid Motion

Section 6 Steady, 2D, potential flow



- 1, Potential flow function
- 2, Stream function
- 3, Flow net
- 4, Superposition of potential flows



Section 6 Steady, 2D, potential, incompressible flow

Basic equations:

1, 2D, potential flow: $\omega_z = 0$

2, Steady flow: $\frac{\partial u_x}{\partial t} = \frac{\partial u_y}{\partial t} = 0$

3, Incompressible flow: $\rho = C$

Euler equations:

$$\left. \begin{aligned} f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} &= u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \\ f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} &= u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \end{aligned} \right\}$$

Continuity equation:

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0$$



1, Flow potential function

- Potential flow: $\omega_z = 0$, $\frac{\partial u_x}{\partial y} = \frac{\partial u_y}{\partial x}$

$\longleftrightarrow$ equivalent to $u_x dx + u_y dy$ is a total differential $d\phi$

- Cartesian coordinates:

$$u_x dx + u_y dy = d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$\left. \begin{aligned} u_x &= \frac{\partial \phi}{\partial x} \\ u_y &= \frac{\partial \phi}{\partial y} \end{aligned} \right\}$$

ϕ ——— potential flow function or flow potential



- Laplace equation form of potential function

Incompressible, 2D flow:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad \text{或} \quad \nabla^2 \phi = 0$$

$$u_x = \frac{\partial \phi}{\partial x}$$
$$u_y = \frac{\partial \phi}{\partial y}$$

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0$$

- Polar coordinates:

$$d\phi = u_r dr + u_\theta r d\theta$$

$$\left. \begin{aligned} u_r &= \frac{\partial \phi}{\partial r} \\ u_\theta &= \frac{1}{r} \frac{\partial \phi}{\partial \theta} \end{aligned} \right\}$$





Chapter 3 Fundamentals of Fluid Motion

Section 6 Steady, 2D, potential flow



- 1, Potential flow function
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2, Stream function



1, Definition

- (Steady), incompressible, 2D flow
- Cartesian coordinates:

Continuity equation: $\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0 \Rightarrow \frac{\partial u_x}{\partial x} = -\frac{\partial u_y}{\partial y}$

$\Longleftrightarrow -u_y dx + u_x dy$ is a total differential $d\psi$

$$d\psi = -u_y dx + u_x dy = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

$$\left. \begin{aligned} u_x &= \frac{\partial \psi}{\partial y} \\ u_y &= -\frac{\partial \psi}{\partial x} \end{aligned} \right\}$$

ψ — is stream function



- Laplace equation form of stream function

Potential flow: $\frac{\partial u_x}{\partial y} = \frac{\partial u_y}{\partial x}$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\nabla^2 \psi = 0$$

- Polar coordinates:

$$d\psi = u_r r d\theta - u_\theta dr$$

$$\left. \begin{aligned} u_r &= \frac{1}{r} \frac{\partial \psi}{\partial \theta} \\ u_\theta &= -\frac{\partial \psi}{\partial r} \end{aligned} \right\}$$





2, Physical meaning

(1) Contour lines: $\psi(x, y) = C$ is a streamline

$$\because \psi(x, y) = C$$

$$\therefore d\psi = -u_y dx + u_x dy = 0$$

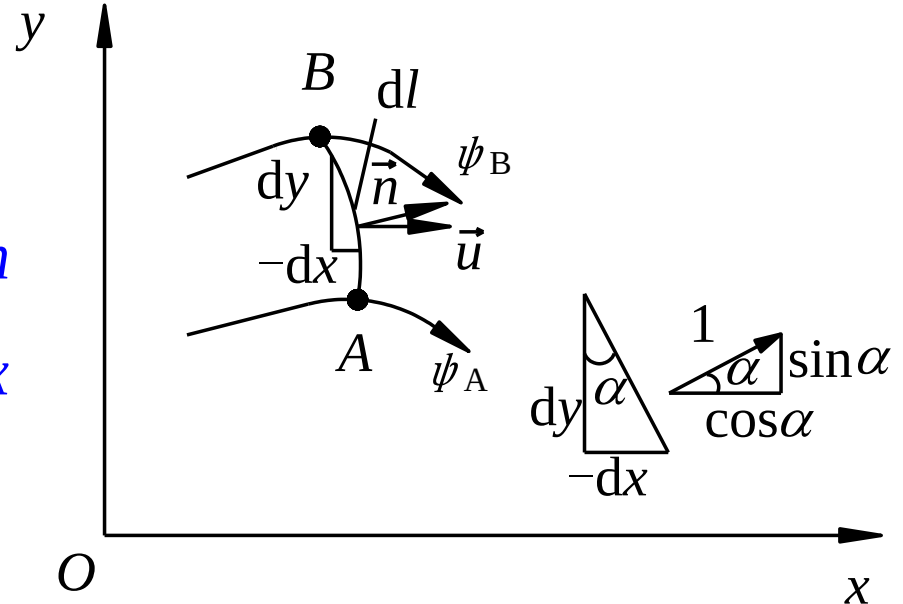
$$\frac{dx}{u_x} = \frac{dy}{u_y}$$



2, In a steady, 2D, incompressible fluid, the difference $d\psi$ between the stream functions of two streamlines is equal to the flow rate per unit width dq passed between the two streamlines.

Flow through the AB-section:

$$\begin{aligned} dq_l &= u_n dn = \vec{u} \cdot \vec{n} dn \\ &= [u_x \cos(n, x) + u_y \cos(n, y)] dn \\ &= u_x dy + u_y (-dx) = u_x dy - u_y dx \\ &= d\psi \end{aligned}$$





From a series of streamlines and their ψ -values, the flow velocity at each point in the flow field can be approximately determined as follows:

$$u_s = \frac{\Delta q}{\Delta A} = \frac{\Delta \psi}{\Delta n}$$

3, The relationship between stream function and flow potential:

$$\begin{cases} \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \\ \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} \end{cases}$$

These are the Cauchy-Riemann equations, i.e., $f(x+iy) = \phi + i\psi$ is a holomorphic (complex differentiable) function



Example 1 2D, point-source flow $u_x = \frac{Cx}{x^2 + y^2}, u_y = \frac{Cy}{x^2 + y^2}, u_z = 0$

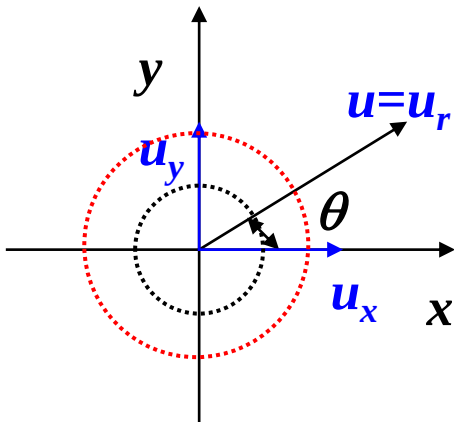
- (1) Is it a potential flow? (2) If so, what is the flow potential φ ?
 (3) Is the flow incompressible? (4) Find the stream function ψ .

$$(1) \quad \frac{\partial u_x}{\partial y} = \frac{-2Cxy}{(x^2 + y^2)^2} \quad \frac{\partial u_x}{\partial y} = \frac{\partial u_y}{\partial x} \quad \text{Yes, potential flow.}$$

$$\frac{\partial u_y}{\partial x} = \frac{-2Cxy}{(x^2 + y^2)^2}$$

$$(2) \quad d\varphi = u_x dx + u_y dy \Rightarrow \varphi = \int \frac{Cx}{(x^2 + y^2)} dx + \frac{Cy}{(x^2 + y^2)} dy$$

Polar coordinates: $u_\theta = 0$



$$u_r = u = \sqrt{u_x^2 + u_y^2} = \sqrt{\frac{C^2(x^2 + y^2)}{(x^2 + y^2)^2}} = \frac{C}{\sqrt{x^2 + y^2}} = \frac{C}{r}$$

$$\therefore \varphi = \int u_x dx + u_y dy = \int \vec{u} \cdot d\vec{r} = \int u_r \cdot dr = \int \frac{C}{r} dr$$

$$= C \ln r + \text{const} = C \ln \sqrt{x^2 + y^2} + \text{const}$$



Example 1 2D, point-source flow $u_x = \frac{Cx}{x^2 + y^2}, u_y = \frac{Cy}{x^2 + y^2}, u_z = 0$

- (1) Is it a potential flow? (2) If so, what is the flow potential ϕ ?
(3) Is the flow incompressible? (4) Find the stream function ψ .

(3) $\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = \frac{C(y^2 - x^2)}{(x^2 + y^2)^2} + \frac{C(x^2 - y^2)}{(x^2 + y^2)^2} = 0$ Yes, incompressible flow

(4) $d\psi = -u_y dx + u_x dy$

$$\psi = \int -u_y dx + u_x dy = \int -\frac{Cy}{x^2 + y^2} dx + \frac{Cx}{x^2 + y^2} dy$$

$$= C \int \frac{xdy - ydx}{x^2 + y^2} = C \int \frac{d(\frac{y}{x})}{1 + (\frac{y}{x})^2} = C \tan^{-1} \frac{y}{x} + \text{const} = C\theta + \text{const}$$



Example 2 two flows: (a) $u_x=1, u_y=2$; (b) $u_x=4x, u_y=-4y$.

(1) Is there a stream function ψ for (a)? If so, what is it?

(2) Is there a flow potential function ϕ for (b)? If so, what is it?

$$(1) \quad \frac{\partial u_x}{\partial x} = 0, \frac{\partial u_y}{\partial y} = 0$$

$$\therefore \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0 \quad \text{Yes, there is a stream function.}$$

$$d\psi = u_x dy - u_y dx = -2dx + dy = d(y - 2x)$$

Stream function: $\psi = y - 2x + C_1$

$$(2) \quad \left. \begin{aligned} \therefore \frac{\partial u_x}{\partial y} &= \frac{\partial(4x)}{\partial y} = 0 \\ \frac{\partial u_x}{\partial x} &= \frac{\partial(-4y)}{\partial y} = 0 \end{aligned} \right\} \therefore \frac{\partial u_x}{\partial y} = \frac{\partial u_x}{\partial x} = 0 \quad \text{Yes, there is a flow potential.}$$

$$d\phi = u_x dx + u_y dy = 4x dx + (-4y) dy = d(2x^2 - 2y^2)$$

Potential flow function: $\phi = 2x^2 - 2y^2 + C_2$



Example 3 Stream function: $\psi = ax^2 - ay^2$;

(1) Is the flow irrotational? (2) If so, find the flow potential.

(3) Prove that the streamlines are orthogonal to the equipotential lines.

$$(1) \quad u_x = \frac{\partial \psi}{\partial y} = -2ay$$

$$u_y = -\frac{\partial \psi}{\partial x} = -2ax$$

$$\therefore \omega_x = \frac{1}{2} \left(\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z} \right) = 0$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x} \right) = 0$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y} \right) = \frac{1}{2} (-2a + 2a) = 0$$

Yes, potential (irrotational) flow.



Example 3 Stream function: $\psi = ax^2 - ay^2$;

(1) Is the flow irrotational? (2) If so, find the flow potential.

(3) Prove that the streamlines are orthogonal to the equipotential lines.

$$(2) \quad \because u_x = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = -2ay$$

(1)

$$\therefore \phi = \int \frac{\partial \psi}{\partial y} dx = -\int 2ay dx = -2axy + C(y)$$

$$u_y = \frac{\partial \phi}{\partial y} = -2ax + C'(y) = -2ax$$

$$\phi = -2axy + C$$





Example 3 Stream function: $\psi = ax^2 - ay^2$;

(1) Is the flow irrotational? (2) If so, find the flow potential.

(3) Prove that the streamlines are orthogonal to the equipotential lines.

(3) Streamline:

$$d\psi = u_x dy - u_y dx = 0$$

Streamline slope:

$$m_1 = \frac{dy}{dx} = \frac{u_y}{u_x}$$

Equipotential line:

$$d\phi = u_x dx + u_y dy = 0$$

Equipotential line slope:

$$m_2 = \frac{dy}{dx} = -\frac{u_x}{u_y}$$

$$\therefore m_1 \times m_2 = -1;$$

Yes, orthogonal.





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3, Flow net

Flow net: mesh formed by streamlines and equipotential lines

1, Nature of flow net

- Streamlines are orthogonal to equipotential lines

Equipotential lines: $\phi(x,y)=C$

$$d\phi = u_x dx + u_y dy = 0$$

$$\left. \frac{dy}{dx} \right|_{\phi = \text{const}} = -\frac{u_x}{u_y}$$

Streamlines: $\psi(x,y)=C$

$$d\psi = -u_y dx + u_x dy = 0$$

$$\left. \frac{dy}{dx} \right|_{\psi = \text{const}} = \frac{u_y}{u_x}$$

Orthogonal: $\left. \frac{dy}{dx} \right|_{\phi = \text{const}} \cdot \left. \frac{dy}{dx} \right|_{\psi = \text{const}} = -1$



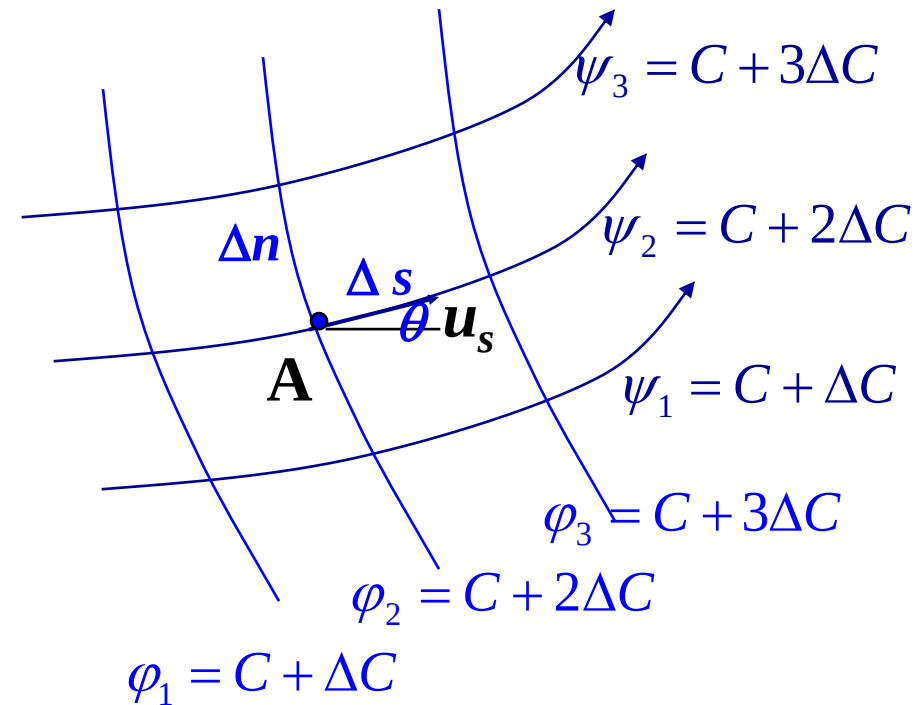
- The ratio of the mesh side lengths of the flow net is equal to the ratio of the increments $\Delta\phi/\Delta\psi$, if $\Delta\phi=\Delta\psi$, the flow mesh is a square grid.

Proof: Flow value difference ΔC , streamline interval Δn , equipotential line interval Δs

$$u_s = \frac{\Delta q}{\Delta A} \approx \frac{\Delta\psi}{\Delta n} = \frac{\Delta C}{\Delta n}$$

$$\text{且 } u_s = \frac{\partial\phi}{\partial s} \approx \frac{\Delta\phi}{\Delta s} = \frac{\Delta C}{\Delta s}$$

$$\therefore \Delta s = \Delta n$$





2, Drawing of flow net

- Graphical method:**

(1) At the solid boundary, the flow velocity component perpendicular to the boundary is zero, so the solid boundary itself is one of the streamlines. Equipotential lines are orthogonal to the boundary.

(2) Make $\Delta C = \text{constant}$.

(3) Draw streamlines and equipotential lines according to a preselected grid scale. Then, the grid is repeatedly modified according to the characteristics of the flow net (try to make each grid drawn into a curved square).



3, Application of flow nets

Flow nets has been widely used to determine the velocity field and pressure field in ideal fluid potential flow, such as seepage.

Velocity field: Because in the flow network, the $\Delta\psi$ between any two adjacent streamlines is the same, the flow rate in the grid $\Delta q = \text{constant}$. Hence, within each grid, the flow velocity is inversely proportional to the spacing Δn .

$$u_s \approx \frac{\Delta q}{\Delta n} \quad \frac{u_1}{u_2} = \frac{\Delta n_2}{\Delta n_1}$$

Flow rate at one point $\implies$ flow rate at another point

Pressure:
$$\frac{p_1 - p_2}{\rho g} = \frac{\Delta p}{\rho g} = z_2 - z_1 + \frac{u_2^2 - u_1^2}{2g}$$

Pressure at one point $\implies$ pressure at another point





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4, Superposition of potential currents

Potential flow superposition: The superposition of several potential flow and stream functions are still potential flow and stream function.

$$\varphi = \sum \varphi_i; \quad \psi = \sum \psi_i$$



Example 1 Superposition of uniform flow and point source flow

Uniform flow: $u_x = u_0$ (常数), $u_y = 0$

Point source flow: $u_r = q / 2\pi r$, $u_\theta = 0$

Uniform flow:

$$d\varphi_1 = u_x dx + u_y dy = u_0 dx \quad \therefore \varphi_1 = u_0 x + C$$

$$d\psi_1 = -u_y dx + u_x dy = u_0 dy \quad \therefore \psi_1 = u_0 y + C$$

This flow net is shown in Figure a.

Point source flow:

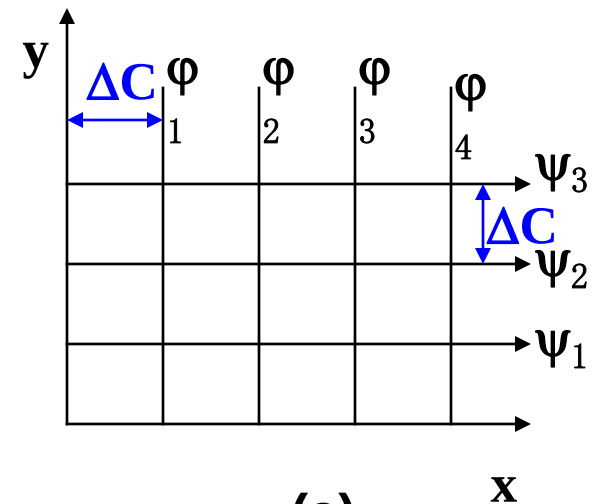
$$d\varphi = u_r dr + u_\theta r d\theta = \frac{q}{2\pi r} dr$$

$$d\psi = u_r r d\theta + u_\theta dr = \frac{q}{2\pi} d\theta$$

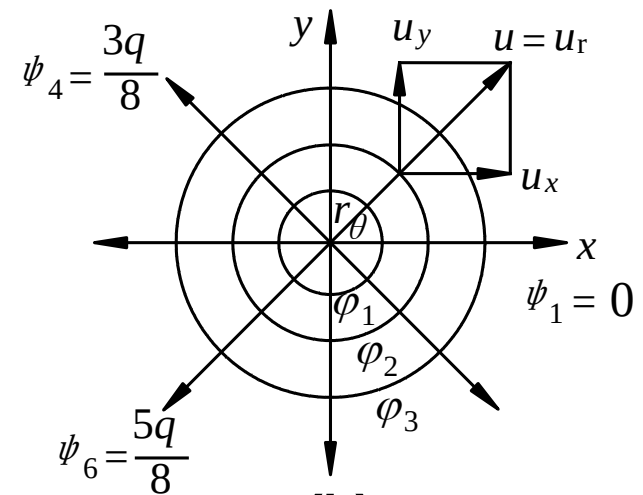
$$\therefore \varphi_2 = \frac{q}{2\pi} \ln r + C$$

$$\therefore \psi_2 = \frac{q}{2\pi} \theta + C$$

Hence, the streamlines are radial lines and the equipotential lines are a cluster of concentric circles orthogonal to the streamlines (Fig. b).



(a)



(b)

The superimposed flow potential function and flow function (Fig. c, next slide)

$$\varphi = \varphi_1 + \varphi_2 = u_0 x + \frac{q}{2\pi} \ln r = u_0 r \cos \theta + \frac{q}{2\pi} \ln r = u_0 x + \frac{q}{2\pi} \ln \sqrt{x^2 + y^2}$$

$$\psi = \psi_1 + \psi_2 = u_0 y + \frac{q}{2\pi} \theta = u_0 r \sin \theta + \frac{q}{2\pi} \theta$$

- Superimposed flow velocity field

$$u_x = \frac{\partial \varphi}{\partial x} = u_0 + \frac{q}{2\pi} \frac{x}{x^2 + y^2} = u_0 + \frac{q}{2\pi} \frac{\cos \theta}{r}$$

$$u_y = \frac{\partial \varphi}{\partial y} = \frac{q}{2\pi} \frac{y}{x^2 + y^2} = \frac{q}{2\pi} \frac{\sin \theta}{r}$$



- Streamline through stagnation point

Stagnation point A: $u_x = 0, u_y = 0$

$$\theta = \pi, r = \frac{q}{2\pi u_0}$$

Streamline: $u_0 r \sin \theta + \frac{q}{2\pi} \theta = C$

or $u_0 y + \frac{q}{2\pi} \theta = C$

At stagnation point: $A(\frac{q}{2\pi u_0}, \pi) \quad C = \frac{q}{2}$

Streamline stagnation point:

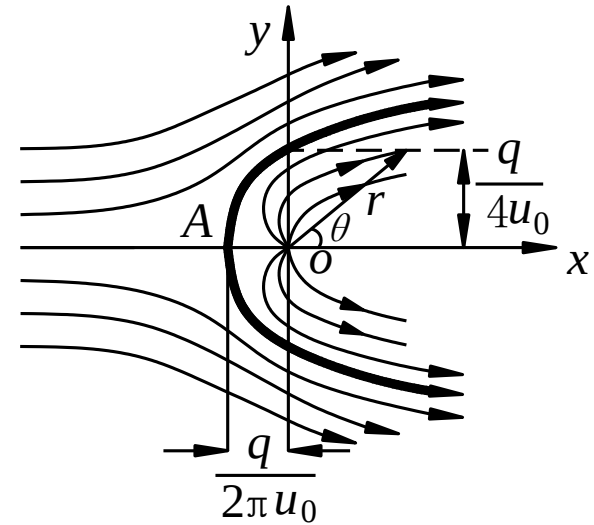
$$u_0 r \sin \theta + \frac{q}{2\pi} (\theta - \pi) = 0$$

$$r = \frac{q}{2\pi u_0} \frac{\pi - \theta}{\sin \theta}$$

$$y = r \sin \theta = \frac{q}{2\pi u_0} (\pi - \theta) = \frac{q}{2\pi u_0} (\pi - \tan^{-1} \frac{y}{x})$$

For $x \rightarrow \infty$ 时, $r \rightarrow \infty, \theta \rightarrow 0, y \rightarrow \frac{q}{2u_0}$

For $x \rightarrow 0$ (即 $\theta \rightarrow \frac{\pi}{2}$) 时, $y \rightarrow \frac{q}{4u_0}$



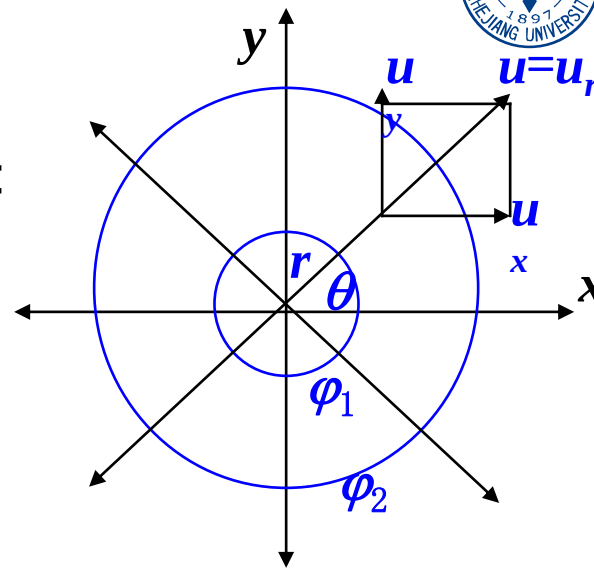
(c)



Example 2 2D point source flow:

$$u_x = \frac{Cx}{x^2 + y^2}; \quad u_y = \frac{Cy}{x^2 + y^2}; \quad u_z = 0$$

Find the pressure distribution.



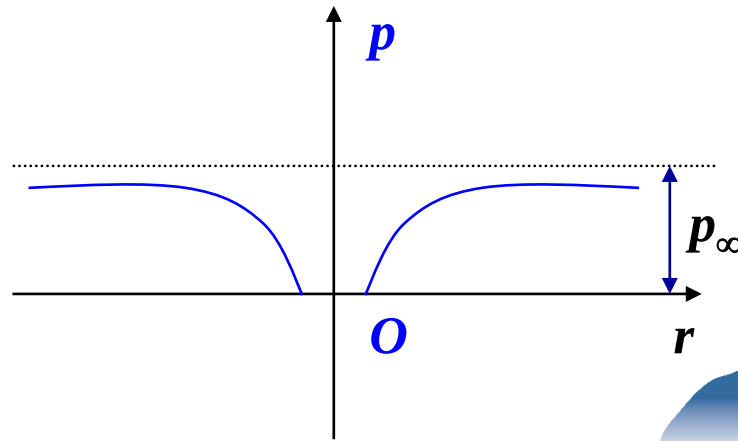
(a)

Bernoulli's equation ($g=0$):

$$\frac{p}{\rho} + \frac{u^2}{2} = C_1$$

$$u = \frac{C}{r}$$

$$p = C' - \frac{\rho}{2} \left(\frac{C}{r} \right)^2$$



(b)

For $r \rightarrow \infty$, $p = p_\infty$, i.e., $C' = p_\infty$

$$p = p_\infty - \frac{\rho}{2} \left(\frac{C}{r} \right)^2$$

Therefore, p is distributed along the r -direction according to the parabolic law, as shown in Figure **b** (previous slide). Finally, the determination of C in the above equation: from the flow rate per unit depth:

$$Q = \int_0^{2\pi} u_r \cdot r d\theta = \int_0^{2\pi} \frac{C}{r} \cdot r d\theta = 2\pi C$$

$$\therefore C = \frac{Q}{2\pi}$$

is called the 2D point source (sink) intensity.





Questions

1, What are the conditions for the existence of the stream function and the potential flow function? Do they all satisfy the Laplace equation?

Stream function: steady, 2D, incompressible flow; potential flow function: steady, 2D, potential flow; if steady, 2D, incompressible, potential flow, both satisfy Laplace equation.

2, what is the physical meaning of the stream function?

(1) Stream function contours, $\psi(x, y) = C$, are streamlines.

(2) In a 2D, incompressible flow, the difference between the flow functions of any two streamlines, $d\psi$, is equal to the flow rate per unit width, dq , that passes between the two streamlines.





3, What is a flow net? What are some of the properties of flow nets? What are the applications?

Flow Net: An orthogonal grid composed of streamlines and equipotential lines describing 2D, steady, incompressible, potential flow.

Properties of the flow net:

- 1) The streamlines and equipotential lines are orthogonal everywhere
- 2) The ratio of the side lengths of each grid in the flow network is equal to the ratio of the differences $\Delta \phi / \Delta \psi$, if we take $\Delta \phi = \Delta \psi$, then the flow net is a square grid.

Flow nets have been widely used to determine the velocity and pressure field of ideal fluids, such as seepage.



Chapter summary



I. Basic concepts

1, Basic concepts and their properties

The nature of the streamlines:

- a, Different streamlines at the same time do not intersect.
- b, Streamlines are smooth curves.
- c, The density of a streamline cluster reflects the flow velocity magnitude.
- Stream function: $\psi(x, y)=C$ is a streamline, $d\psi=dq$
- Flow net: orthogonal, requires steady, 2D, incompressible potential flow.
- Bernoulli's equation: $gz + \frac{p}{\rho} + \frac{u^2}{2} = C'$

Continuity equation:
$$\frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} + \frac{\partial(\rho u_z)}{\partial z} + \frac{\partial \rho}{\partial t} = 0$$

Incompressible CE:
$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0$$





Rearranged:

$$* \quad z + \frac{p}{\rho g} + \frac{u^2}{2g} = C$$

physical meaning

geometric meaning

z — Specific potential energy

Elevation head

$\frac{p}{\rho g}$ — Specific pressure energy

Pressure head

$\frac{u^2}{2g}$ — Specific kinetic energy

Flow head

$z + \frac{p}{\rho g}$ — Total specific potential energy

Hydraulic (piezometric) head

$z + \frac{p}{\rho g} + \frac{u^2}{2g}$ — Total specific energy

Total head

Applicable conditions:

a: ideal fluid, mass force only gravity, steady, incompressible, potential flow

b: ideal fluid, mass force only gravity, steady, incompressible, along streamlines



2, Two methods of describing fluid motion (Lagrange's and Euler's method)

Eulerian acceleration:

$$\text{1D flow: } a = \frac{du}{dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial s}$$

$$\text{3D flow: } a_x = \frac{du_x}{dt} = \frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z}$$

$\frac{\partial u}{\partial t}$ ——local acceleration $\frac{\partial u}{\partial t} = 0$, steady flow

$u \frac{\partial u}{\partial s}$ ——convective acceleration $u \frac{\partial u}{\partial s} = 0$, uniform flow



3, Flow classification

Flow pattern { laminar
turbulent

ordered
disorganized

Time evolution { Steady flow: $\frac{\partial A}{\partial t} = 0$ ($A = u, p, \rho, h \dots$)
Unsteady flow: $\frac{\partial A}{\partial t} \neq 0$

Spatial evolution { Uniform flow $\frac{\partial \vec{u}}{\partial \vec{s}} = 0$ 即 $\frac{\partial u}{\partial s} = 0$ 且 $\frac{u^2}{r} = 0$ }
Non-uniform flow { Gradually-varying flow
Rapidly-varying flow }

On the same flow cross-section:
$$z + \frac{p}{\rho g} = C$$



By spatial
coordinates

- 1D flow: flow velocity depends on only one coordinate
- 2D flow: flow velocity depends on two coordinates
- 3D flow: flow velocity depends on all three coordinates

Angular
velocity

- Rotational flow: $\frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y} \neq 0$ 或 $\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z} \neq 0$ 或 $\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x} \neq 0$
- Potential flow: $\frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y} = 0$ 且 $\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z} = 0$ 且 $\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x} = 0$



2, Steady, 2D, potential flow

- Stream function: $d\psi = -u_y dx + u_x dy = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$

$$\frac{\partial \psi}{\partial y} = u_x, \frac{\partial \psi}{\partial x} = -u_y, d\psi = dq_l$$

Existence criterion: steady, 2D, incompressible flow

Streamlines: $d\psi = -u_y dx + u_x dy = 0, \psi = \text{const}$

- Potential flow function: $d\phi = u_x dx + u_y dy = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$

$$\frac{\partial \phi}{\partial x} = u_x, \frac{\partial \phi}{\partial y} = u_y$$

Existence criterion: steady, 2D, potential flow





- Relationship between stream function and potential flow function:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

$$\nabla^2 \phi = 0, \nabla^2 \psi = 0$$

$$\phi = \phi_1 + \phi_2 + \cdots, \psi = \psi_1 + \psi_2 + \cdots$$

The superposition of potential flows is a potential flow.

